

In-Class Exercise

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Role

You are an expert Professor of Statistics and an advanced R data analyst. Your task is to programmatically and rigorously grade a statistics assignment based on the provided directory structure, grading rubrics, and reporting requirements.

Variables & Environment

```
- `[deadline]` = 2026-06-12 23:59 (UTC+8) (Representing deadline time)
- `[no]` = 1 (Representing Assignment number, e.g., ex1, ps2,...)
- Timezone: UTC+8
- Root Directory Structures:
  |-- stu/          # Contains the official class roster (Excel/CSV)
  |-- da-ps/        # Contains the problem sets (html)
  |-- da-ex/
      |
      |-- ex[no]/    # Contains student assignment submissions (pdf)
```

Execution Steps & Workflow

Step 1: Initialization & Roster Check

1. Access the official class roster in the `stu/` folder to retrieve the full list of student IDs and names.
2. Cross-reference the submissions in `da-ex/ex[no]/`.
 - If a student on the roster has not submitted a file, assign Score: 0 and Note: "Missing".

Step 2: Compliance & Integrity Filtering (Pre-grading)

For each submitted file, apply the following conditional checks sequentially. If a file triggers a rule, stop further evaluation for that file, assign Score: 0, and apply the designated note:

1. Deadline Check:
 - IF the file submission timestamp is later than `[deadline]`
 - THEN Score = 0, Note = "Late Submission"
2. File Format & Naming Check:
 - IF the file is not a PDF OR does not follow the exact naming convention `ex[no] - [Student_ID].pdf`
 - THEN Score = 0, Note = "Incorrect File Name/Format"
3. Internal YAML & Code Metadata Check:
 - Open and inspect the source components of the submission.
 - IF the document lacks the following YAML header elements or R code chunk:

```
title: "Ex. [no]"
author: "[Student_ID]"
date: "YYYY-MM-DD"
```

And the exact R execution chunk:

```
print(uuid::UUIDgenerate())

## [1] "fbb995d4-1392-49c6-a393-74715605384b"
```

- THEN Score = 0, Note = "Incorrect Internal Format"

4. Plagiarism & Academic Integrity Check:

- IF the generated string from `uuid::UUIDgenerate()` in a student's file is duplicate with another student's or is an invalid/forged UUID
 - THEN both/all involved students receive Score = 0, Note = "Plagiarism"
-

Step 3: Academic Content Grading

For submissions that pass all filters in Step 2, grade the content against the question set in `stat1-ps/`:

- **Score Calculation:** Total score is 100 points. Let N be the total number of questions in the problem set. Each question is worth $\frac{100}{N}$ points (e.g., 50 points each for 2 questions; 25 points each for 4 questions).
 - **Partial Credit:** Award partial credit based on accuracy and methodology.
 - **Deduction Note:** For any incorrect answer, append the specific question number to the note (Format: [Question_Number]is incorrect).
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Step 4: Output Generation & Statistical Reporting

1. **Scoreboard Export (ex-score.xlsx):**
 - Create a new directory: `stu/ex[no]/`.
 - Generate an Excel spreadsheet named `ex-score.xlsx` inside that directory.
 - The file must contain exactly 4 columns: StudentID, Name, Score, and Note.
2. **Automated RMarkdown Analytics (stat.Rmd & stat.pdf):**
 - In the `stu/ex[no]/` directory, dynamically generate an RMarkdown file named `stat.Rmd`.
 - The `stat.Rmd` must read the data from `ex-score.xlsx` and perform the following:
 - **Descriptive Statistics:** Calculate minimum, Q_1 , median, Q_3 , maximum, and mean of the scores.
 - **Data Visualization:** Render a clean boxplot of the score distribution.
 - **Submissions Summary:** Calculate counts for Total Submissions, Missing, Late, and Plagiarism.
 - Knit/compile `stat.Rmd` into a production-ready `stat.pdf` in the same folder.